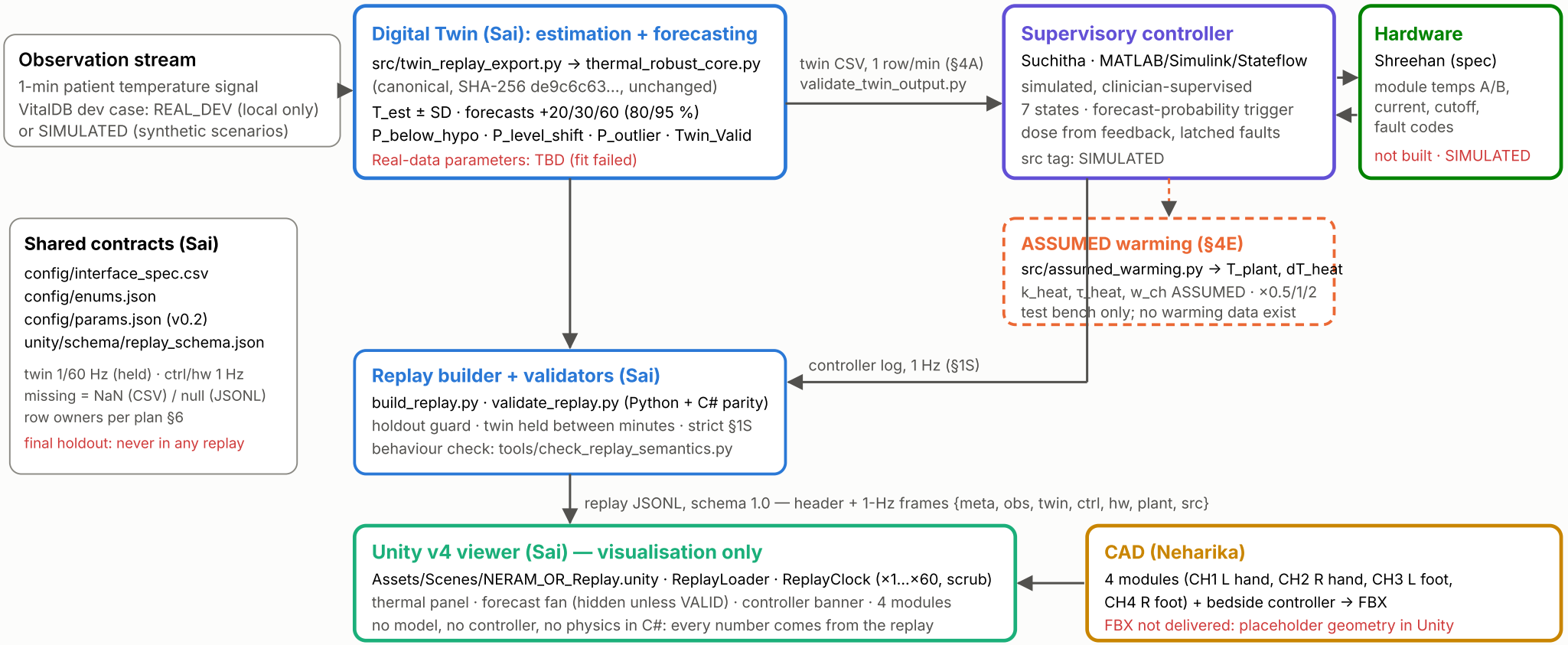
NERAM — system architecture (open-loop code submission, 20 Oct 2026)

A sequentially updated, patient-specific thermal digital twin for intraoperative temperature forecasting (proof-of-concept). Owner of this document: Sai. Component blocks: controller — Suchitha; hardware — Shreehan; CAD — Neharika.

Proof-of-concept. SIMULATED / ASSUMED PROTOTYPE RESPONSE where marked. Input = "patient temperature signal" (VitalDB Solar8000/BT, anatomical site not established; not core temperature). Controller and hardware are simulated; the warming effect is ASSUMED; no clinical claim. The real-data twin fit has not succeeded yet.



Component boundaries

Block	Owner	Status at 17 Oct
Digital Twin export	Sai	Implemented; runs the canonical filter on SYNTHETIC data with ASSUMED parameters. Real-data fit FAILED (quantised data, empty learning windows); correction pending approval.
Replay builder, validators, schema	Sai	Implemented and tested (62-fixture parity corpus, JSON Schema, behaviour checks).
Unity viewer	Sai	v4 project tested in the Editor and as a Windows build on 10 Oct; 11 synthetic replays.
Supervisory controller	Suchitha	Logs TC05/TC07/TC13 received; none commands heating; TC07 shows the over-temperature latch. Detailed state diagram: Suchitha's block.
Hardware	Shreehan	Specification only; fault_codes.csv and thresholds TBD. Block diagram: Shreehan's block.
CAD	Neharika	FBX export pending; Unity uses named placeholders.

Assumed warming equation (plan §4E)

$$d_t = \Sigma\, w_i \cdot Heat_i/100;\quad h_t = h_{t-1} + (\Delta t/\tau_{heat})(d_t - h_{t-1});$$
$$\Delta T_t = \Delta T_{t-1} + k_{heat} \cdot (\Delta t/60) \cdot h_t;\quad T_{plant} = T_{pass} + \Delta T_{heat}$$

ASSUMED, illustrative, not estimated: VitalDB has no warming data. Sensitivity ×0.5/×1/×2 for every use. Closed loop (twin re-filters the warmed trajectory) is summit work and is not implemented.

Source tags shown on every panel

Group	Tag
Observation	REAL_DEV (development case, local only) · SIMULATED (synthetic)
Twin	MODEL (canonical filter) · SIMULATED for v4's stand-in scenarios
Controller	SIMULATED
Hardware	SIMULATED
Plant / warming	ASSUMED

Scenario mapping (plan §4R → replay files)

Plan	Shown as
SCN_A usual care (development case)	Not available (needs the development fit); layout stand-in SYN_SCN_A0, SYNTHETIC
SCN_B predictive warming	SYN_SCN_B (SYNTHETIC, stand-in twin and controller, ASSUMED warming)
SCN_C over-temperature fault	SYN_SCN_C (stand-in) and SYN_TC07 (Suchitha's TC07 log; latch only, no heating, no reset)
SCN_D twin degraded	SYN_SCN_D (SYNTHETIC stand-in)

Not built / TBD

Real-data twin parameters · development replay DEV_<pseudo> · hardware fault-code display text · CAD FBX · a controller log with heating · summit closed loop · final holdout (one run after preregistration).